CHAPTER 15 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

UAS Law, Ethics, Regulation, Legislation, and Adjudication

UAS Legal Frameworks

Determine which rules apply

Identify the jurisdiction, operating category and airspace restrictions before planning the flight.

Check operator obligations

Verify applicable registration, pilot competency and operational authorization requirements.

Document the decision

Record the planned route, relevant approvals and the person responsible for the flight.

EASA reference: Easy Access Rules for UAS

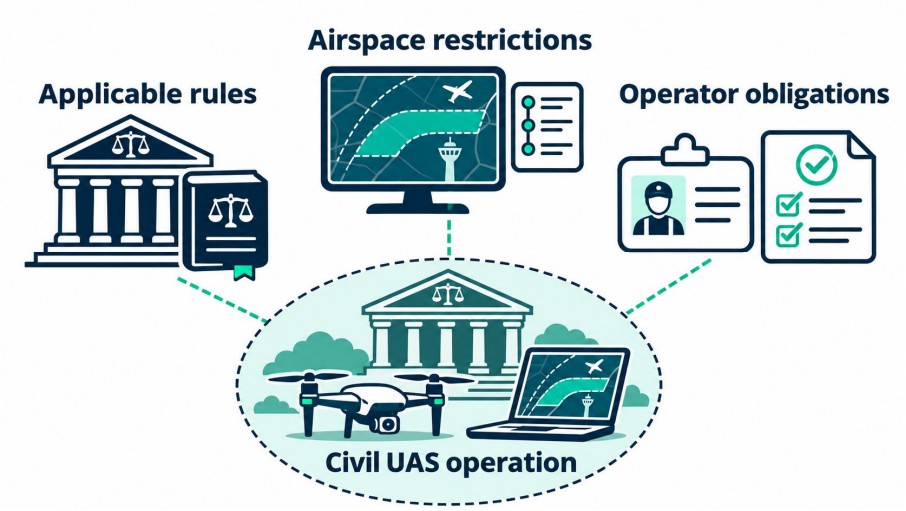


Figure S15.1. Planning a civil UAS operation involves identifying the rules that apply to the mission, checking airspace restrictions along the proposed route, and confirming the operator’s obligations before flight.

UAV Remote Identification and Tracking Requirements

Identify the aircraft

Where required, Remote ID broadcasts identification and location information to nearby receivers.

Interpret the track

Record time, receiver coverage and gaps when reconstructing movement from received messages.

Verify flight authorization

Cross-check the observed identifier and track against applicable authorization records.

EASA reference: Specific Category and Remote Identification

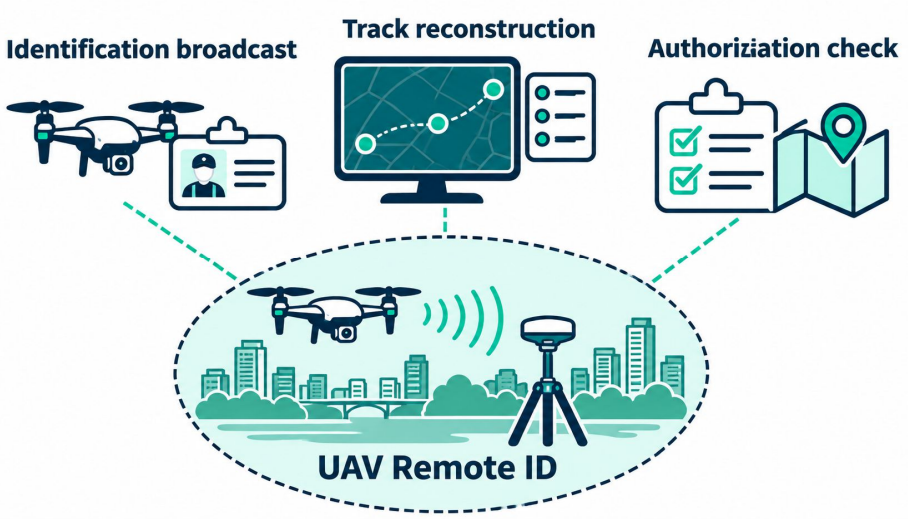


Figure S15.2. A ground receiver captures a drone's Remote ID broadcasts. The received position reports can help reconstruct its flight path, while flight authorization is checked separately against applicable records.

UAV Privacy, Data Protection, and Surveillance Concerns

Limit what the sensor collects

Choose routes and camera angles that reduce incidental capture of people and private property.

Control access and retention

Restrict access to raw footage, set a retention period and redact identifying details before sharing.

Document the purpose

Record why the data are needed and check applicable privacy and data protection obligations.

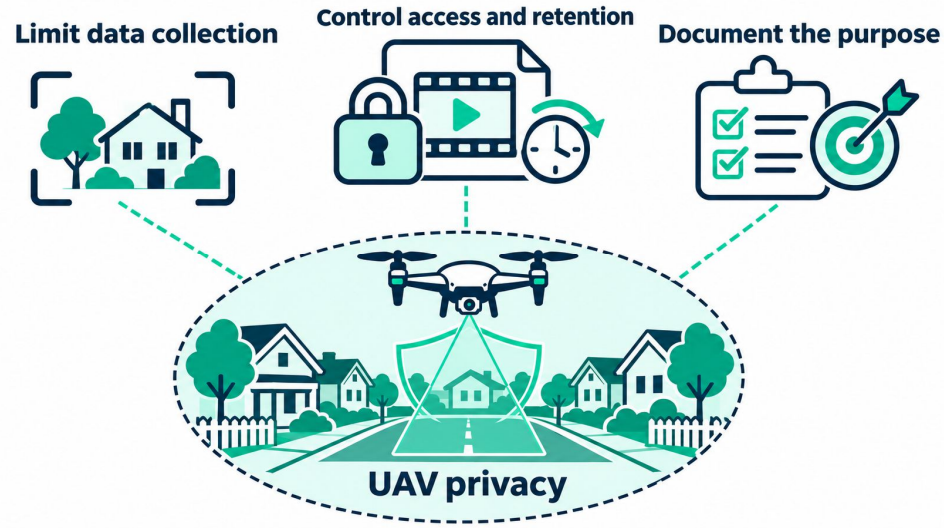


Figure S15.3. A UAV operating near homes illustrates three privacy safeguards: document why imagery is needed, limit the camera's field of view to reduce incidental recording, and define who can access the footage and how long it will be kept.

Ethics and Regulation of Drone Cybersecurity Research with Human Subjects

Seek an ethics determination

Ask the institutional review body whether the study requires review before recruiting participants.

Protect participants

Explain the study, risks and withdrawal options in the consent process, where required.

Limit collection and access

Collect only necessary data and store participant identifiers separately from research records.

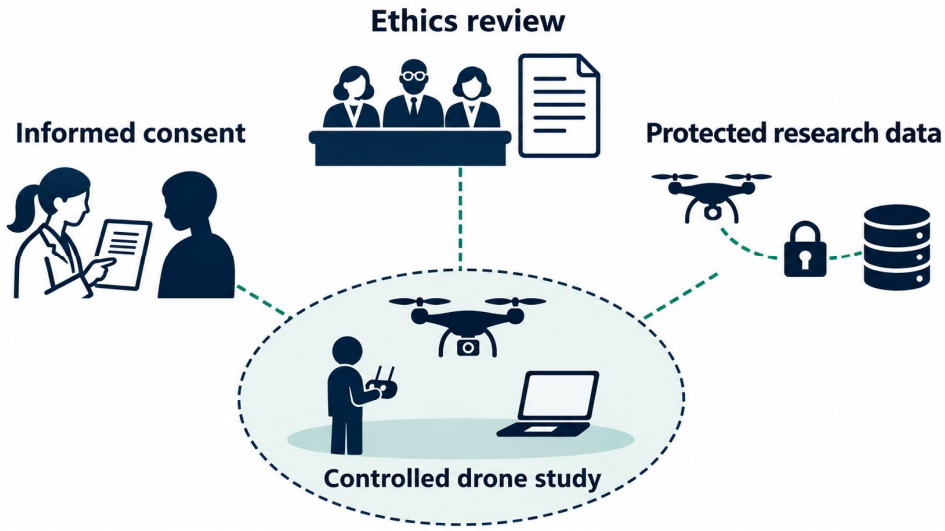


Figure S15.4. A controlled drone cybersecurity study with human participants, showing how ethics review, informed consent, and safeguards for research data support the study.

Adjudication of UAS Incidents and Cyber–Physical Violations

Preserve original evidence

Secure flight logs, telemetry and operator records. Document each transfer in the chain of custody.

Reconstruct the timeline

Align timestamps and corroborate events across independent sources. Record gaps and uncertainty.

Support adjudication

Identify the applicable rule and distinguish verified observations from investigator inferences.

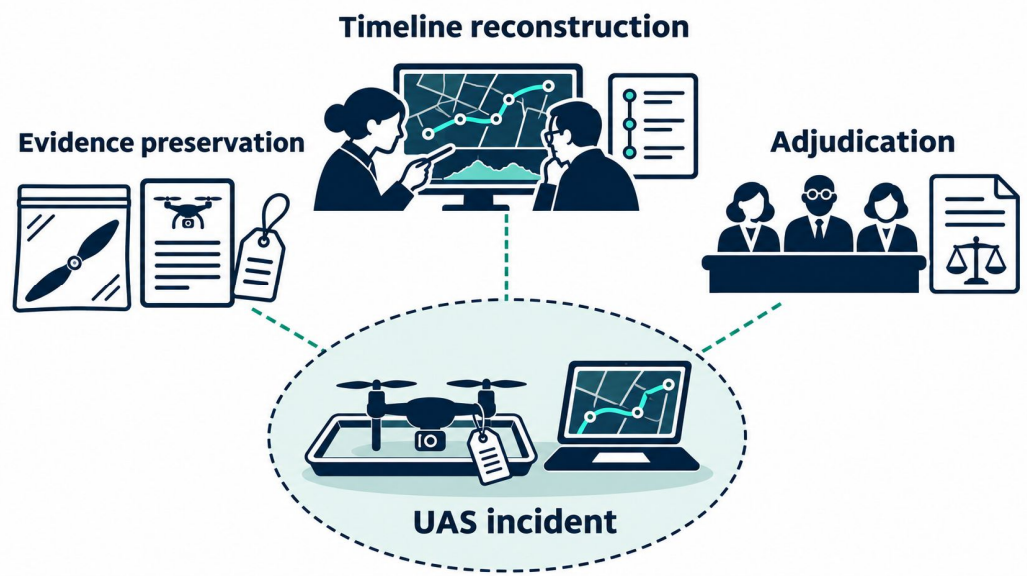


Figure S15.5. Investigators preserve physical evidence and digital flight records, then align timestamps to reconstruct the incident. The adjudicating body evaluates those findings against the applicable rules.

Regulatory Challenges for Autonomous UAV Operations

Define the operating limits

Specify the weather, airspace and communications conditions under which the aircraft can operate.

Assign human oversight

Define who monitors the flight, when they intervene and how they can command a safe fallback.

Test and record failures

Exercise lost-link and degraded navigation scenarios. Retain results and configuration records.

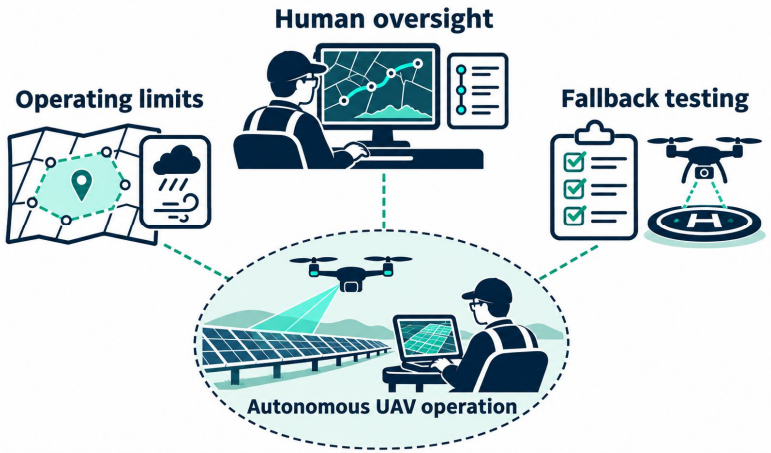


Figure S15.6. An autonomous UAV inspects a solar farm under remote human supervision. The illustration highlights three regulatory considerations: defining operating limits, assigning responsibility for intervention, and testing fallback procedures.